

STRUCTURE & FUNCTIONS OF CEREBELLUM & ITS CONNECTIONS

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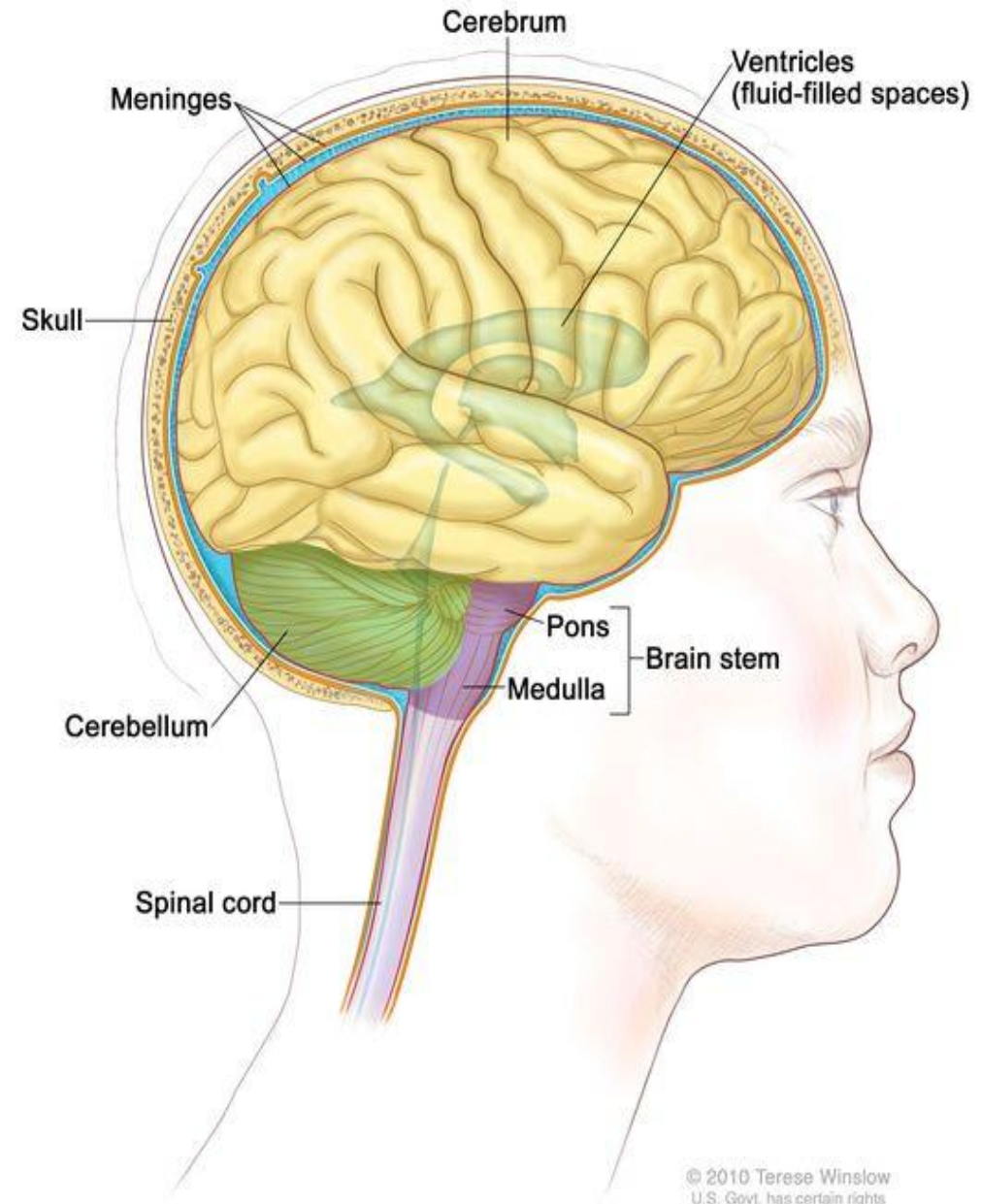
LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- Identify the location & major external & internal structures of cerebellum.
- Describe anatomical divisions of cerebellum(lobes, hemispheres, vermis, cortex & nuclei)
- Explain input(afferent) & output(efferent) pathways in relation to functional divisions of cerebellum.
- Describe the major functions of cerebellum.
- Identify the effects of cerebellar damage in clinical practice.

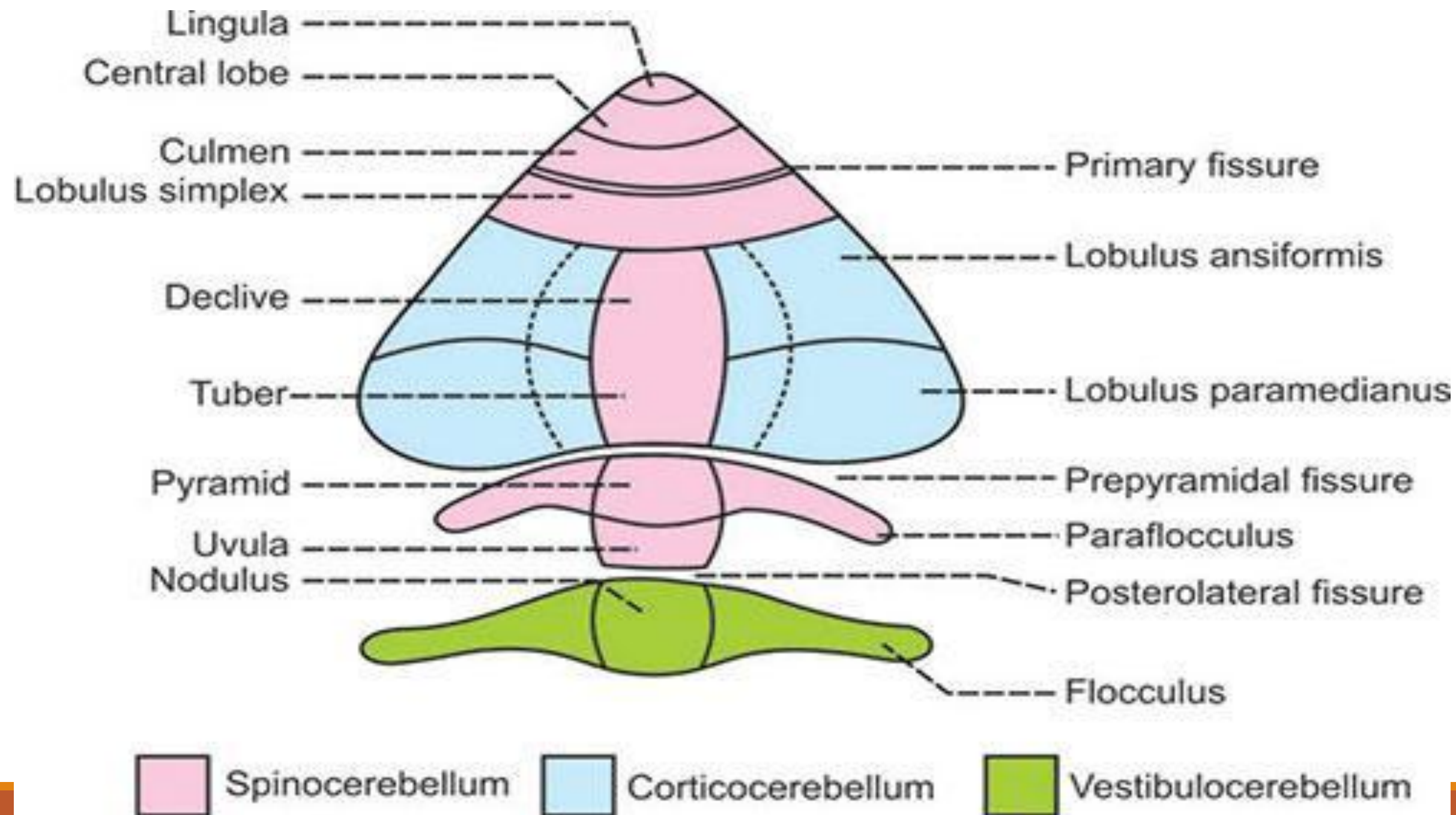
INTRODUCTION

- Cerebellum (means ***little brain***) is a major structure of hindbrain located at the base of skull behind the brainstem.
- It is situated at the very bottom of the brain below the occipital lobe & just above the point where the neck meets the spine.
- It controls balance, coordinates voluntary movements & aids in motor learning.
- Cerebellum is shaped like a half-circle around the brainstem & consists of a narrow worm-like central body called ***vermis*** & two lateral lobes, the right & left ***cerebeller hemispheres***.



CENTRAL PART OF CEREBELLUM 'THE VERMIS'

- Vermis is formed by nine parts, the part of vermis on the upper surface of cerebellum is called ***superior vermis*** & the part of vermis on the lower surface of cerebellum is called ***inferior vermis***.
- Nodulus, a part of inferior vermis extends on either sides as an elongated & lobulated structure called ***flocculus***.
- Nodulus & flocculi together constitute the ***flocculonodular lobe*** of cerebellum.
- Extensions of pyramid on either sides are termed as ***paraflocculus***.



FISSURES ON THE SURFACE OF VERMIS

Three fissures are present over the surface of vermis:

Primary fissure: present between culmen & lobulus simplex.

Prepyramidal fissure: present between tuber & pyramid.

Posterolateral fissure: present between uvula & nodulus.

CEREBELLAR HEMISPHERES

□ Cerebellar hemispheres are the extended portions on either side of vermis.

□ Each hemisphere consists of two portions:

Lobulus ansiformis(ansiform lobe): constitute the larger portion of cerebellar hemisphere.

Lobulus paramedianus(paramedian lobe): constitute the smaller portion of cerebellar hemisphere.

DIVISIONS OF CEREBELLUM

Cerebellum can be divided into different parts on the basis of three methods:

- ❖ Anatomical divisions
- ❖ Phylogenetic divisions
- ❖ Physiological or functional divisions.

ANATOMICAL DIVISION

On structural basis, the cerebellum can be divided into three portions i.e. the ***anterior lobe***, the ***posterior lobe*** & the ***flocculonodular lobe***.

1. Anterior lobe

- Lingula, central lobe & culmen constitute the anterior lobe of cerebellum.
- Anterior lobe of cerebellum is separated from the posterior lobe by primary fissure.

2. Posterior lobe

- Posterior lobe of cerebellum consists of lobulus simplex, declive, tuber, pyramid, uvula, paraflocculi & the two lobes of hemispheres i.e. the ansiform lobe & the paramedian lobe.

3. Flocculonodular lobe

- It includes nodulus & the lateral extensions on either side called flocculus.
- Flocculonodular lobe is separated from the rest of cerebellum by posterolateral fissure.

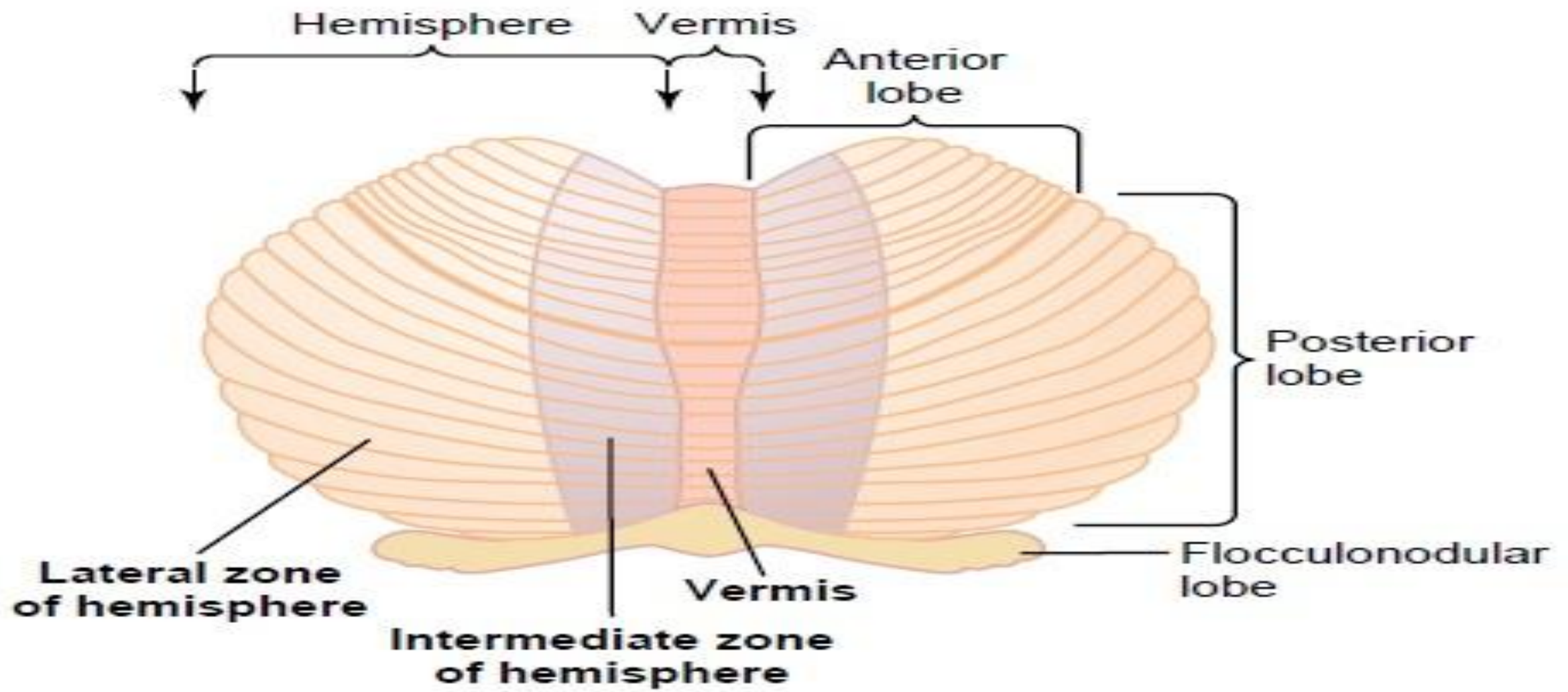


Figure 56-2

Functional parts of the cerebellum as seen from the posteroinferior view, with the inferiormost portion of the cerebellum rolled outward to flatten the surface.

PHYLOGENETIC DIVISION

Phylogenetically, the cerebellum can be divided into two main divisions:

1. Paleocerebellum
2. Neocerebellum.

Paleocerebellum

Paleocerebellum is the phylogenetically oldest part of cerebellum.

It consists of two divisions, ***archicerebellum*** which includes flocculonodular lobe & ***paleocerebellum proper*** which includes lingula, central lobe, culmen, lobulus simplex, pyramid, uvula & paraflocculi.

Neocerebellum

Neocerebellum is the phylogenetically newer portion of cerebellum.

It consists of declive, tuber & the two portions of cerebellar hemispheres i.e. ansiform lobe & paramedian lobe.

PHYSIOLOGICAL DIVISION

Based upon functions, cerebellum is divided into three divisions:

1. Vestibulocerebellum
2. Spinocerebellum
3. Corticocerebellum

Vestibulocerebellum

Vestibulocerebellum includes flocculonodular lobe which forms the archicerebellum.

Spinocerebellum

Spinocerebellum includes lingula, central lobe, culmen, lobulus simplex, declive, tuber, pyramid, uvula & paraflocculi & medial portions of lobulus ansiformis & lobulus paramedianus.

Corticocerebellum

It includes lateral portions of lobulus ansiformis & lobulus paramedianus.

FUNCTIONAL ANATOMY OF CEREBELLUM

□ Cerebellum is made up of outer **gray matter** called **cerebellar cortex** & an inner **white matter**.

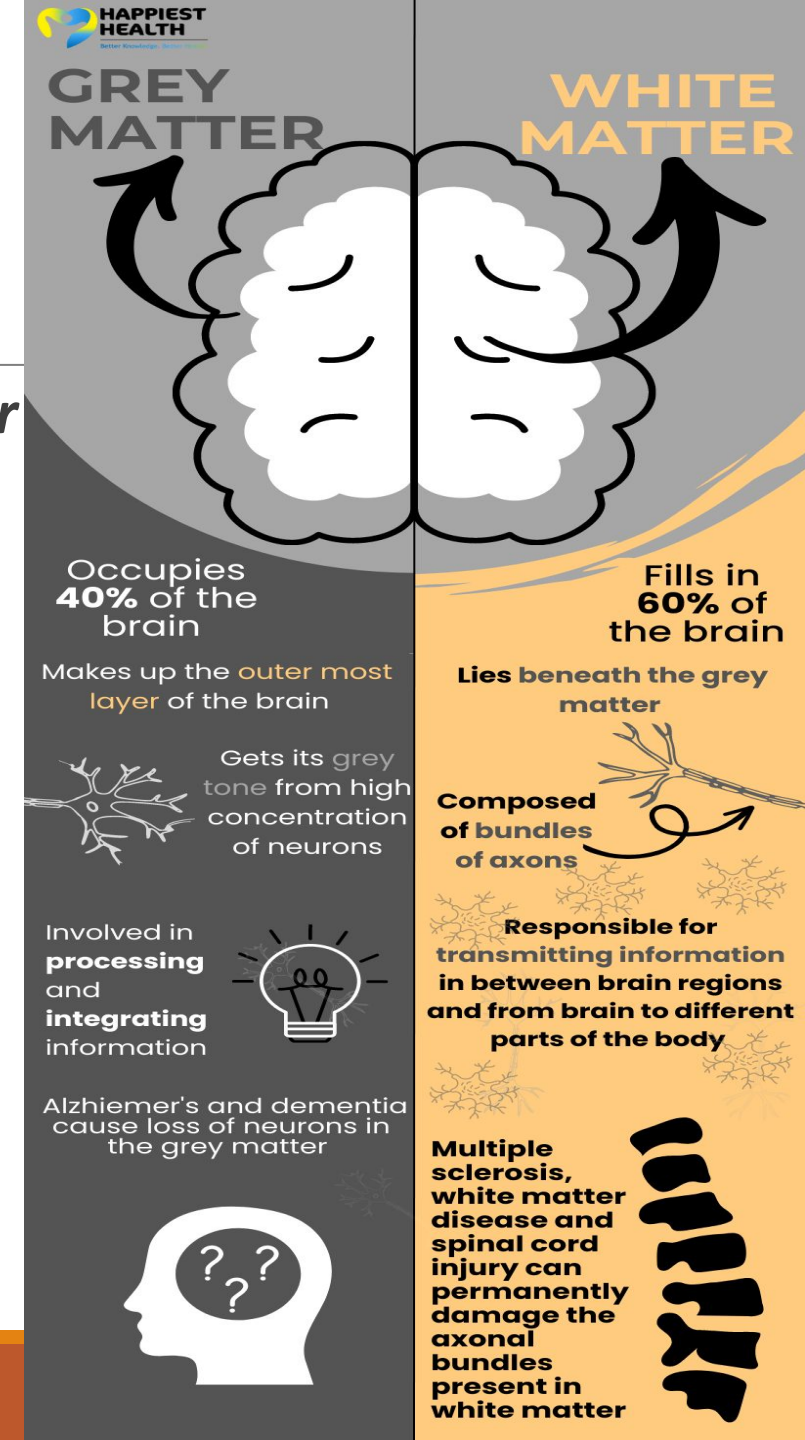
□ White matter is formed by afferent & efferent nerve fibers of cerebellum.

□ Gray masses called cerebellar nuclei are present within the white matter.

GRAY MATTER

□ Gray matter is made up of structures arranged in three layers

1. Outer molecular or plexiform layer
2. Intermediate Purkinje layer
3. Inner granular layer



1. MOLECULAR OR PLEXIFORM LAYER

□ It is the outermost layer of cerebellar cortex & contains cells arranged in two strata.

□ ***Superficial stratum*** contains few star-shaped cells called ***stellate cells***.

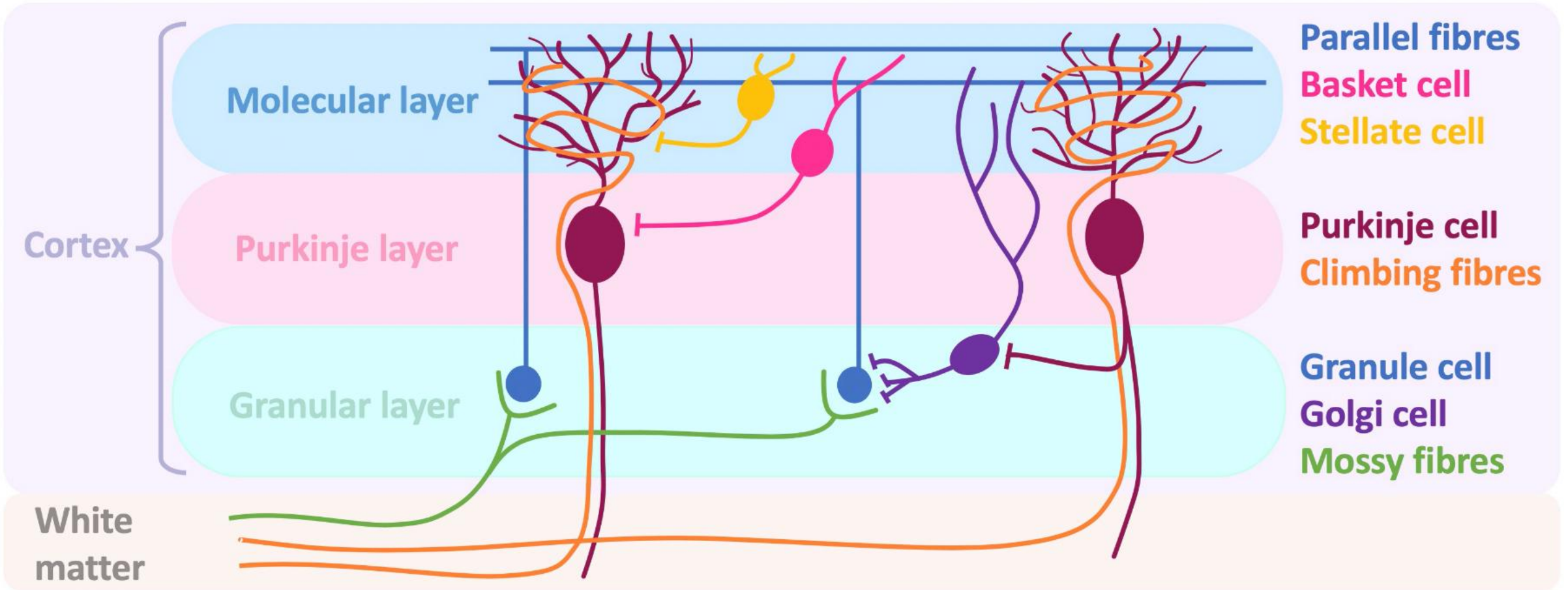
□ ***Deep stratum*** contains ***basket cells***.

□ In addition to the stellate & basket cells, following structures are also present in the molecular layer:

- I. **Parallel fibers:** Parallel fibers are the axons of granule cells present in the granular layer.
- II. **Terminal portions of climbing fibers** (afferent fibers from medulla).
- III. **Dendrites of Purkinje cells & golgi cells.**

CELL JUNCTIONS IN MOLECULAR LAYER

- Dendrites of stellate cells & basket cells synapse with the parallel fibers which are the axons of granule cells.
- Axons of stellate cells synapse with the dendrites of Purkinje cells.
- Axon of basket cells descends down into the Purkinje layer & forms the transverse fiber that synapse with the soma of Purkinje cells.
- Dendrites of Purkinje cells synapse with the climbing fibers & parallel fibers.
- Dendrites of golgi cells present in the inner granular layer enter the molecular layer & synapse with the parallel fibers.



2. INTERMEDIATE OR PURKINJE LAYER

- Purkinje layer is the thinnest layer situated between the outer molecular layer & inner granular layer.
- This layer consists of a single layer of Purkinje cells (largest neurons in the body).
- Purkinje cells are termed as ***final common path of cerebellar cortex*** because impulses from different parts of cerebellar cortex are transmitted to other parts of brain only through Purkinje cells.
- Dendrites of Purkinje cells ascend through the entire thickness of molecular layer & synapse either with the climbing fibers or the parallel fibers.
- Axons of basket cells form the transverse fibers which descend down & terminate on the soma of Purkinje cells.
- Axons of Purkinje cells descend into the white matter & end on the cerebellar nuclei & vestibular nuclei via cerebellovestibular tract.

GRANULAR LAYER

- Innermost layer of cerebellar gray matter present between Purkinje layer & the cerebellar white matter.
- It is formed by interneurons called ***granule cells*** & ***golgi cells***.
- Axons of granule cells ascend into the molecular layer to form parallel fiber & synapse with dendrites of Purkinje cells, stellate cells, basket cells & golgi cells.
- Dendrites of granule cells & the axon & few dendrites of golgi cell synapse with ***Mossy fiber***.
- The synaptic area of these cells called ***glomerulus*** is encapsulated by the processes of glial cells.

AFFERENT FIBERS TO CEREBELLAR CORTEX

□ Cerebellar cortex receive afferent signals through two types of fibers, climbing fibers & mossy fibers.

1. CLIMBING FIBERS

□ These fibers originate from neurons of inferior olivary nucleus present in the medulla & reach cerebellum through olivocerebellar tract.

□ Inferior olivary nucleus receive proprioceptive signals from different parts of body through spinal cord & vestibular system & transmits these signals to the cerebellar cortex via climbing fibers.

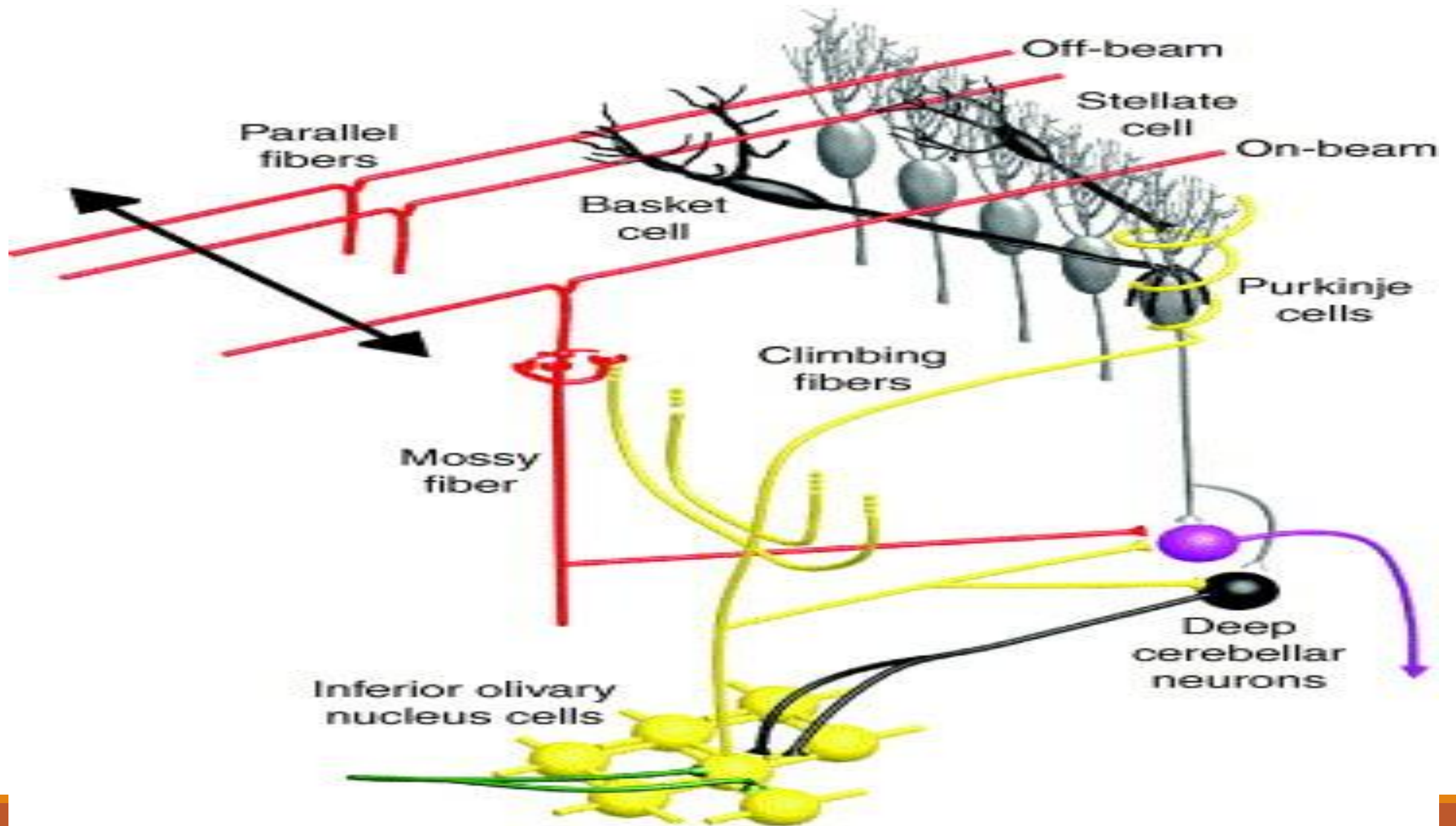
□ After reaching the cerebellum, climbing fibers ascend into the molecular layer & end on dendrites of Purkinje cells.

□ As climbing fibers of olivocerebellar tract send collaterals to cerebellar nuclei so impulses are also conveyed to cerebellar nuclei via climbing fibers in addition to the cerebellar cortex.

□ Each climbing fiber innervates a single Purkinje cell.

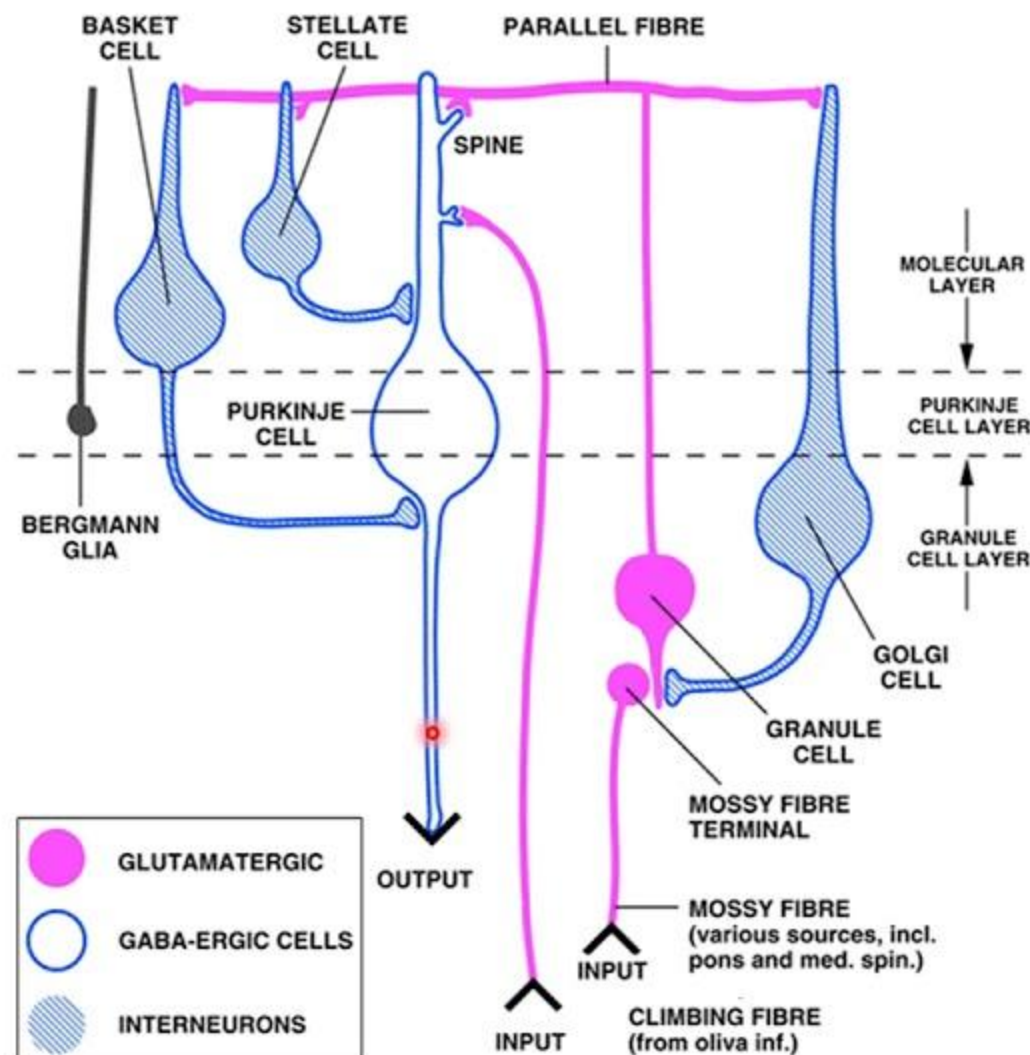
2. MOSSY FIBERS

- Mossy fibers originate from many sources like motor areas of cerebral cortex, pons, medulla & spinal cord & some mossy fibers also arise from cerebellar nuclei.
- Mossy fibers arising from these areas send collaterals to cerebellar nuclei while reaching the cerebellar cortex so like climbing fibers Mossy fibers also transmit afferent impulses to cerebellar nuclei in addition to cerebellar cortex.
- Mossy fibers reach the granular layer & divide into many terminals.
- Each terminal enters a specialized structure called **glomerulus** & ends in a large expanded structure forming the central portion of glomerulus.
- Dendrites of granule cells & axon & dendrites of golgi cells synapse on the mossy fiber giving a **thick bushy appearance** like **moss plant** hence named mossy fibers.



Input/output cells

- Input
 - Climbing fibers
 - Mossy fibers
- Output: Purkinje Cells
 - In Purkinje cell layer
 - The **only** projections that exit the **cerebellar cortex**
 - Forms **inhibitory synapses (GABA)**



NEURONAL ACTIVITY IN CEREBELLAR CORTEX & NUCLEI

Neuron	Action on	Action	Neurotransmitter
Climbing fibers	Purkinje cells and Cerebellar nuclei	Excitation	Aspartate
Mossy fibers	Granule cells, Golgi cells, and Cerebellar nuclei	Excitation	Glutamate
Granule cells	Purkinje cells, Stellate cells, and Basket cells	Excitation	Glutamate/Aspartate
Stellate cells	Purkinje cells	Inhibition	GABA
Basket cells	Purkinje cells	Inhibition	GABA
Golgi cells	Granule cells	Inhibition	GABA
Purkinje cells	Cerebellar nuclei, Vestibular nuclei	Inhibition	GABA
GABA = Gama aminobutyric acid			

CEREBELLAR NUCLEI

□ Cerebellar nuclei are the masses of gray matter scattered in the white matter of cerebellum.

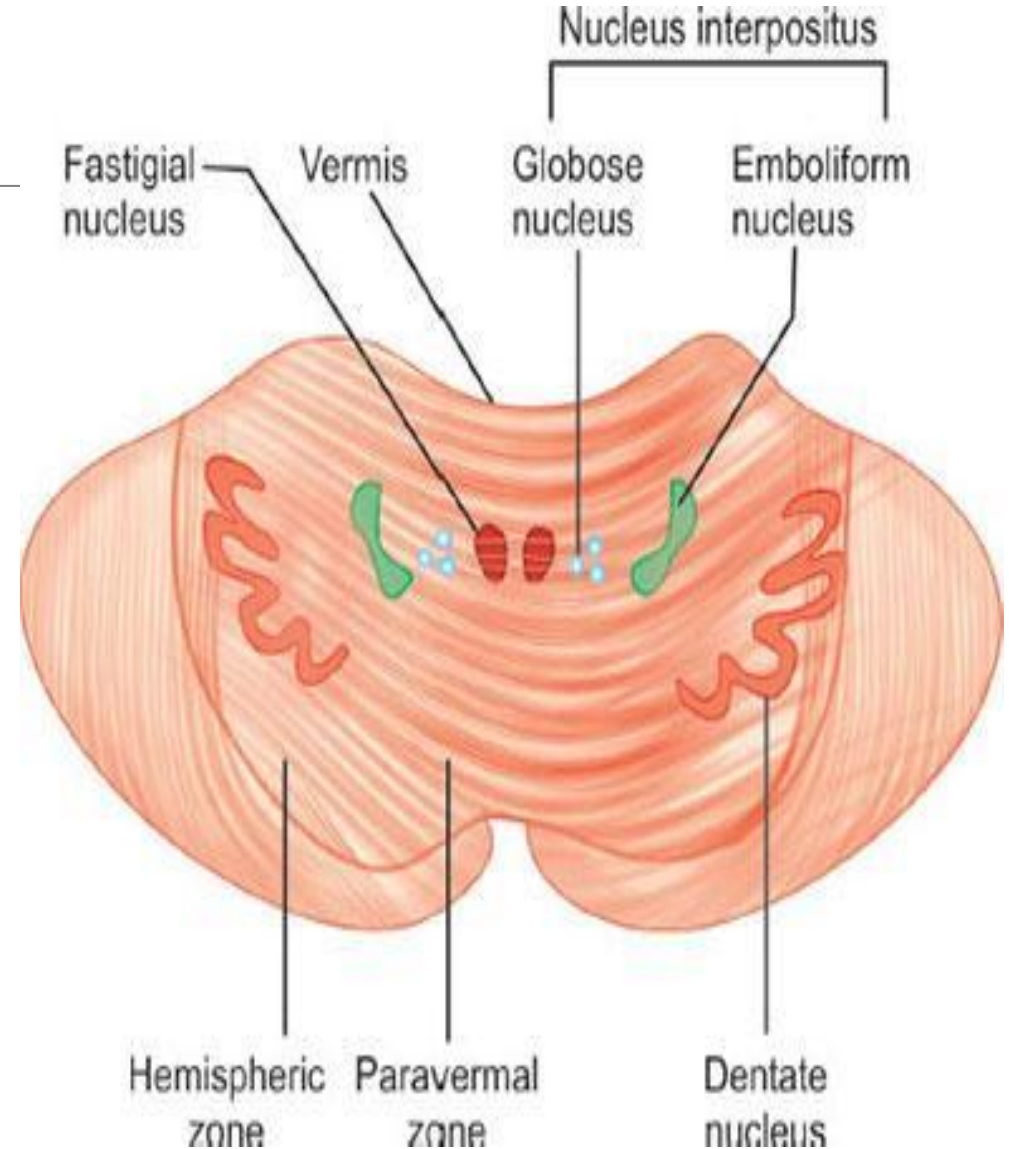
□ Four cerebellar nuclei are present on either side namely:

Fastigial nucleus: placed near the midline on the roof of ventricle IV.

Globosus nucleus: present lateral to nucleus fastigi.

Emboliform nucleus: situated below nucleus fastigi & nucleus globosus.

Dentate nucleus: largest cerebellar nucleus situated most laterally.



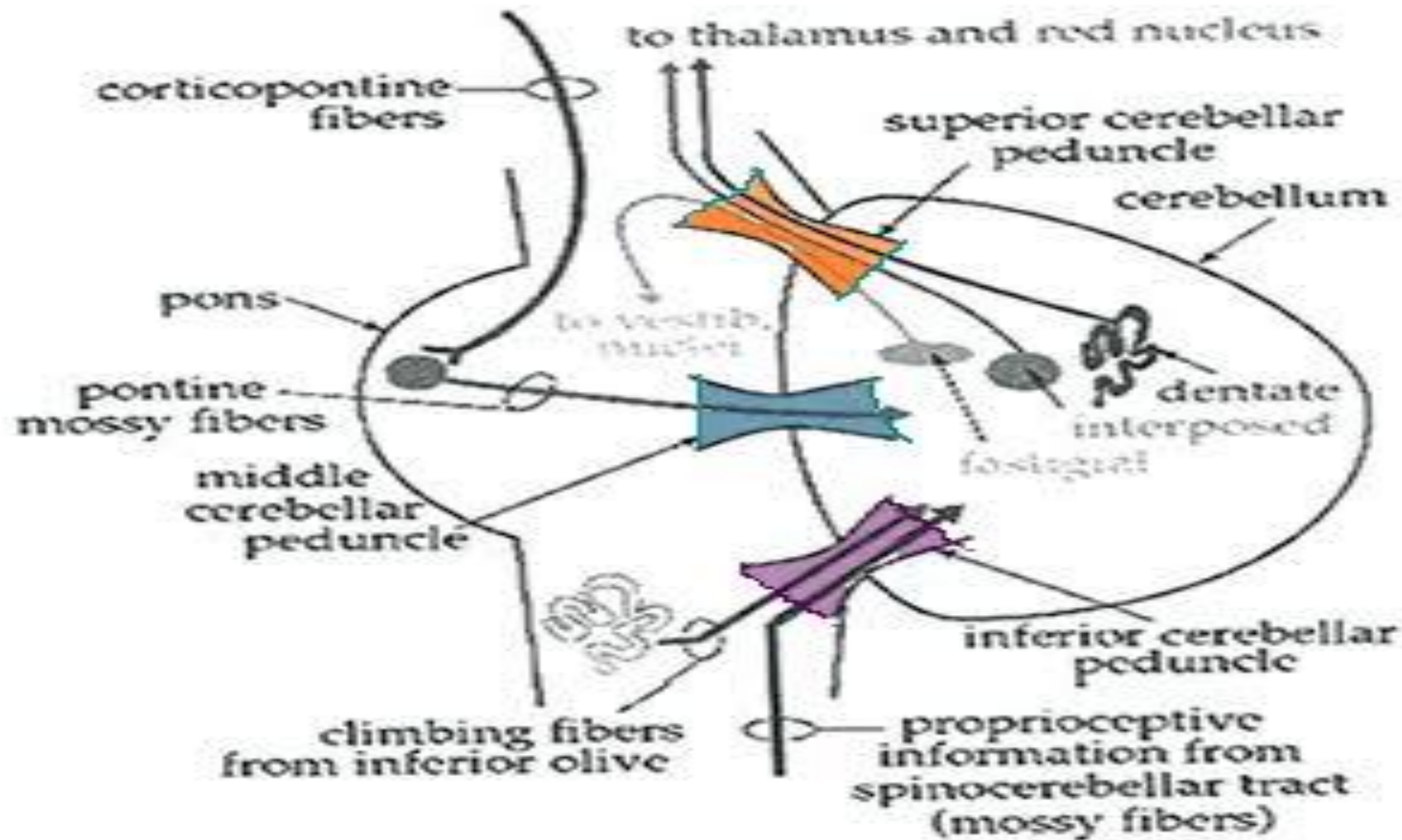
WHITE MATTER OF CEREBELLUM

□ White matter of cerebellum is formed by afferent & efferent nerve fibers classified into three groups:

- 1) **Association fibers:** connect different regions of same cerebellar hemisphere.
- 2) **Commissural fibers:** connect both halves of cerebellar hemisphere.
- 3) **Projection fibers:** connect cerebellum with other parts of CNS.

□ Projection fibers are arranged in three bundles:

- I. **Inferior cerebellar peduncles:** contain predominantly afferent fibers that transmit impulses from tactile receptors, proprioceptors & receptors present in the vestibular apparatus of cerebellum.
- II. **middle cerebellar peduncles(brachia pontis):** contain predominantly afferent fibers most of which are commissural fibers that connect both halves of cerebellar cortex.
- III. **Superior cerebellar peduncles(brachia conjunctivae):** contain predominantly efferent fibers.



VESTIBULOCEREBELLUM (ARCHICEREBELLUM)

- Vestibulocerebellum is the oldest part of cerebellum phylogenetically hence called as ***archicerebellum***.
- It includes flocculonodular lobe formed by ***nodulus*** of vermis & its lateral extensions ***flocculi***.
- Vestibulocerebellum maintains posture & equilibrium.

CONNECTIONS OF VESTIBULOCEREBELLUM

AFFERENT CONNECTIONS

Vestibulocerebellar tract

- ❑ Formed by fibers arising from vestibular nuclei situated in medulla & pons.
- ❑ It passes through the inferior cerebellar peduncle of same side & reaches cerebellar nuclei, nucleus globosus, nucleus emboliformis & nucleus fastigi.
- ❑ Fibers from these nuclei reach the vestibulocerebellum.
- ❑ Vestibular nuclei receive nerve fibers from the vestibular apparatus present in the inner ear through vestibular division of VIII cranial nerve (cochlear nerve).

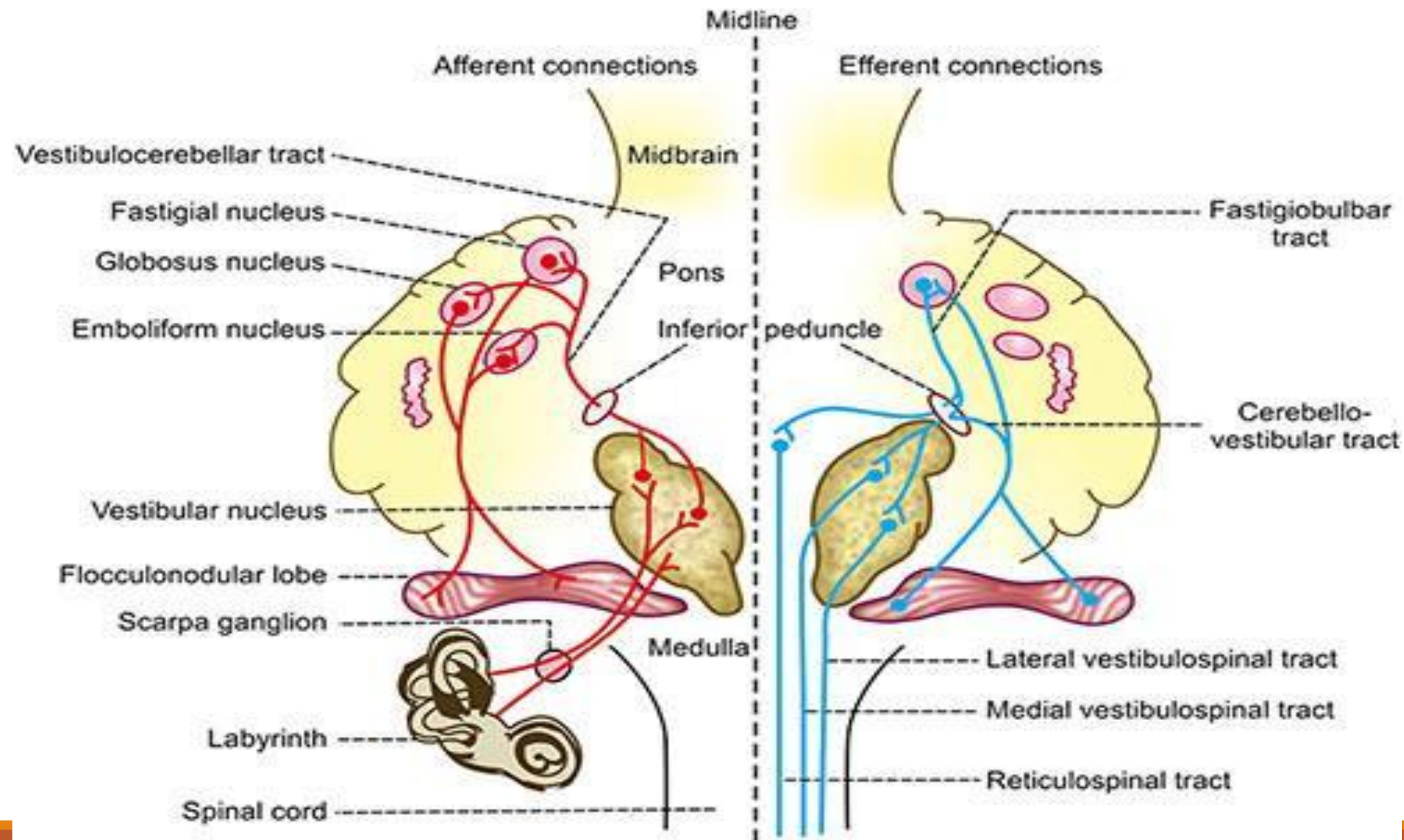
EFFERENT CONNECTIONS

Cerebellovestibular tract

- Fibers arise from flocculonodular lobe & run through inferior cerebellar peduncle of same side & end on the vestibular nuclei situated in the brainstem.
- Fibers from vestibular nuclei form medial & lateral vestibulospinal tracts & terminate on the medial group of alpha motor neurons in the spinal cord.

Fastigiobulbar tract

- Fibers arise from fastigial nucleus pass through inferior cerebellar peduncle of same side & end on vestibular nuclei & reticular formation in medulla oblongata.
- Fibers of vestibulospinal tract arising from vestibular nuclei terminate on alpha motor neurons in the spinal cord.
- Fibers of reticulospinal tract arise from reticular formation & terminate on gamma motor neurons in spinal cord.



FUNCTIONS OF VESTIBULOCEREBELLUM

- Vestibulocerebellum regulates ***muscle tone, posture & equilibrium*** by receiving impulses from the vestibular apparatus.
- Lesions of vestibulocerebellum result in reduced muscle tone(hypotonia) & failure to maintain posture & equilibrium.

SPINOCEREBELLUM (PALEOCEREBELLUM)

- Spinocerebellum is connected with the spinal cord & forms major receiving area of cerebellum for sensory inputs.
- It includes *medial portions of cerebellar hemisphere, paraflocculi* & parts of vermis i.e. *lingula, central lobe, culmen, lobulus simplex, declive, tuber, pyramid & uvula*.
- Spinocerebellum is associated with the maintenance of muscle tone & anticipatory adjustment of muscle contraction during movement.

CONNECTIONS OF SPINOCEREBELLUM

AFFERENT CONNECTIONS

Dorsal spinocerebellar tract

- Fibers of this tract arise from Clarke's column of cells in the dorsal gray horn of spinal cord & reaches spinocerebellum through inferior peduncle of same side.
- This tract conveys proprioceptive information regarding position & movements from limbs of same side.

Ventral spinocerebellar tract

- Fibers of this tract arise from marginal cells in the dorsal gray horn of spinal cord.
- After originating, fibers cross the midline & ascend in opposite side to reach spinocerebellum through superior cerebellar peduncle.
- This tract conveys proprioceptive signals to spinocerebellum regarding position & movements of opposite limbs.

Cuneocerebellar tract

- Fibers arise from accessory cuneate nucleus situated lateral to cuneate nucleus of medulla & reaches spinocerebellum through inferior cerebellar peduncle of same side.
- This tract send proprioceptive impulses to spinocerebellum from upper limb, upper trunk & neck

Olivocerebellar tract

- Olivocerebellar tract is formed by climbing fibers originating from inferior olivary nucleus of medulla.
- The fibers then cross the midline & reach spinocerebellum through inferior cerebellar peduncle of opposite side.
- Inferior olivary nucleus receive afferent fibers from brainstem nuclei of same side, spinal cord through spino-olivary tract of same side & cerebral cortex of opposite side.
- Olivocerebellar tract also gives collaterals to cerebellar nuclei especially to nucleus globosus & nucleus emboliformis.
- This tract sends proprioceptive impulses from the body & also conveys output signals from cerebral cortex to spinocerebellum.

Pontocerebellar tract

- Fibers arise from pontine nuclei, crosses the midline & reaches spinocerebellum through middle cerebellar peduncle of opposite side.
- Pontine nuclei receive afferent impulses cerebral cortex.
- Pontocerebellar tract sends information to spinocerebellum regarding motor signals discharged from cerebral cortex.

Tectocerebellar tract

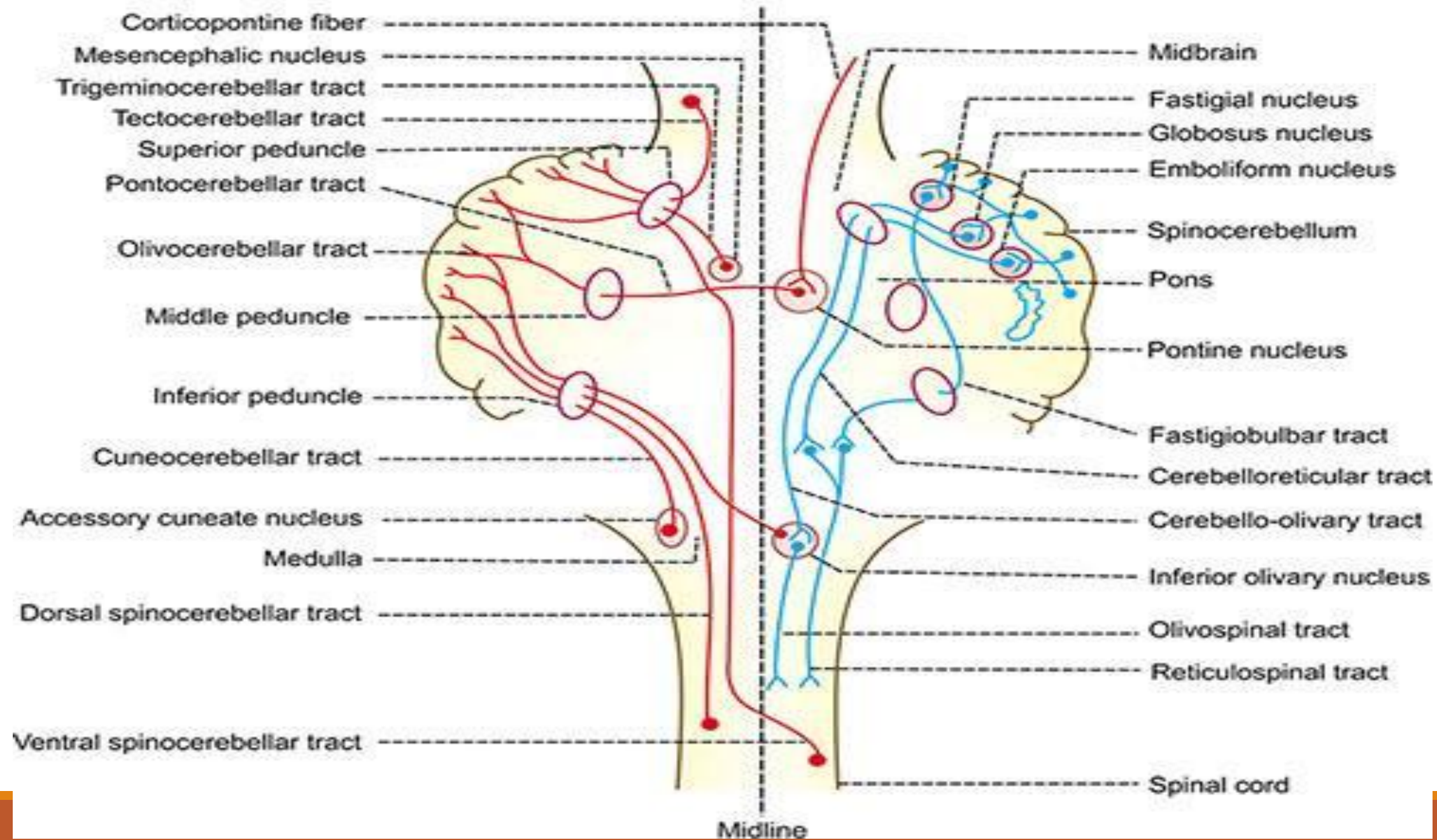
- Fibers arise from superior & inferior colliculi of tectum in midbrain & reaches spinocerebellum through superior cerebellar peduncle of same side.
- Tectocerebellar tract conveys visual impulses from superior colliculus & auditory impulses from inferior colliculus to spinocerebellum.

Trigeminocerebellar tract

- Fibers of this tract originate from mesencephalic nucleus of trigeminal nerve & reaches spinocerebellum through superior cerebellar peduncle of same side.
- This tract transmits proprioceptive signals from jaw muscles & TMJ to spinocerebellum & also the sensory impulses from periodontal tissues to spinocerebellum.

Afferent connections

Efferent connections



EFFERENT CONNECTIONS

Fastigiobulbar tract

□ Fibers arise from fastigial nucleus & passes through superior cerebellar peduncle of same side to reach reticular formation.

Cerebelloreticular tract

□ Fibers arise from nucleus emboliformis & nucleus globosus, pass through superior cerebellar peduncle of same side & end in reticular formation.

□ From reticular formation, reticulospinal tract arise & end on gamma motor neurons of spinal cord.

Cerebello-olivary tract

□ Fibers arise from nucleus emboliformis & nucleus globosus to reach the inferior olivary nucleus of same side through superior cerebellar peduncle.

□ Olivospinal tract arises from olivary nucleus & fibers of this tract terminate on alpha motor neurons of spinal cord.

FUNCTIONS OF SPINOCEREBELLUM

- Spinocerebellum maintains tone, posture & equilibrium by receiving sensory impulses from tactile receptors, proprioceptors, visual receptors & auditory receptors.
- Spinocerebellum is the receiving area for tactile, proprioceptive, auditory & visual impulses & also receives cortical impulses through pontine nuclei.
- Spinocerebellum controls postural reflexes by modifying muscle tone & facilitates discharge from gamma motor neurons of spinal cord.
- Increased discharge from gamma motor neurons increases muscle tone whereas lesion, destruction or abolishing the function of spinocerebellum results in hypotonia & postural disturbances.
- As spinocerebellum also receives impulses from optic & auditory pathway, it helps in adjustment of posture & equilibrium in response to visual & auditory impulses.

CORTICOCEREBELLUM (NEOCEREBELLUM)

- It is the largest part of cerebellum.
- It is called ***corticocerebellum*** or ***cerebrocerebellum*** due to its connection with cerebral cortex.
- It includes lateral portions of cerebellar hemispheres.
- It is associated with planning, programing & coordination of skilled movements.

CONNECTIONS OF CORTICOCEREBELLUM

AFFERENT CONNECTIONS

Pontocerebellar tract

- Largest tract in the body & fibers arise from pontine nuclei, cross the midline & reach corticocerebellum through middle cerebellar peduncle off opposite side.
- It receives signals from motor area of cerebral cortex & transmits them to corticocerebellum, hence called as ***corticopontocerebellar circuit***.

Olivocerebellar tract

- Fibers arise from inferior olivary nucleus in the medulla, crosses midline to reach corticocerebellum via inferior cerebellar peduncle of opposite side & terminate on dentate nucleus & cerebellar cortex.
- Inferior olivary nucleus receives impulses from brainstem, spinal cord & cerebral cortex & transmit these impulses to corticocerebellum via olivocerebellar tract.
- This tract is formed by ***climbing fibers***.

EFFERENT CONNECTIONS

- Efferent signals from corticocerebellum are conveyed mainly via ***dentate nucleus***.
- Fibers from dentate nucleus pass via superior cerebellar peduncle, cross the midline to form decussation with the fibers of opposite side & then divide into two tracts:

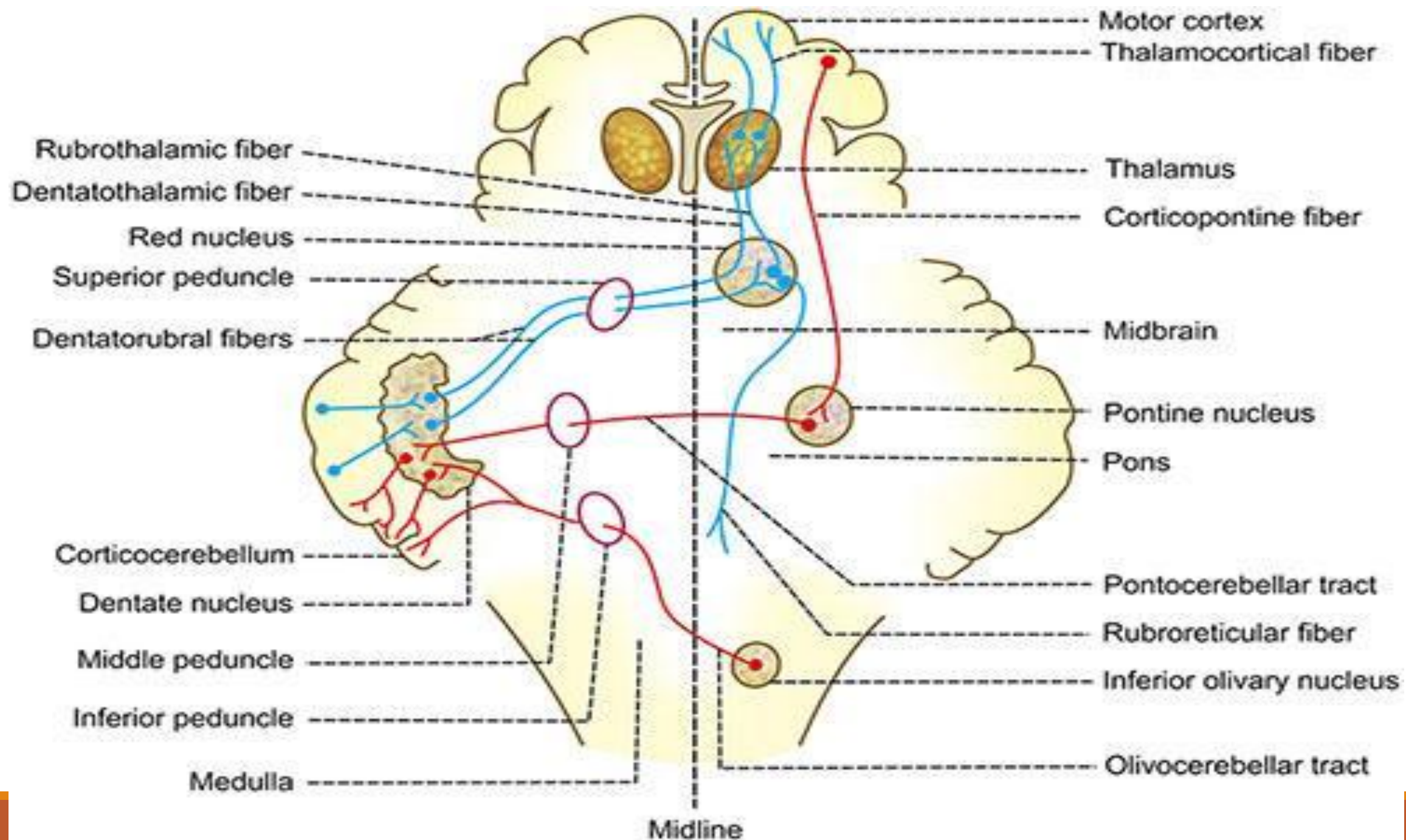
Dentatothalamic tract

- In this tract, fibers pass through ***red nucleus*** without synapsing & terminate in lateral ventral nucleus of thalamus.
- From thalamus, efferent signals then pass through ***thalamocortical fibers*** into motor area of cerebral cortex.

Dentatorubral tract

□ After crossing the midline, fibers terminate in the red nucleus as dentatorubral tract & then project via three tracts:

- I. **Rubrothalamic tract:** originate from red nucleus & terminate in lateral ventral nucleus of thalamus to project into the cerebral cortex via thalamocortical fibers.
- II. **Rubroreticular tract:** descends down to end in reticular formation & projects further into spinal cord via reticulospinal tract.
- III. **Rubrospinal tract:** projects directly into the spinal cord from red nucleus.



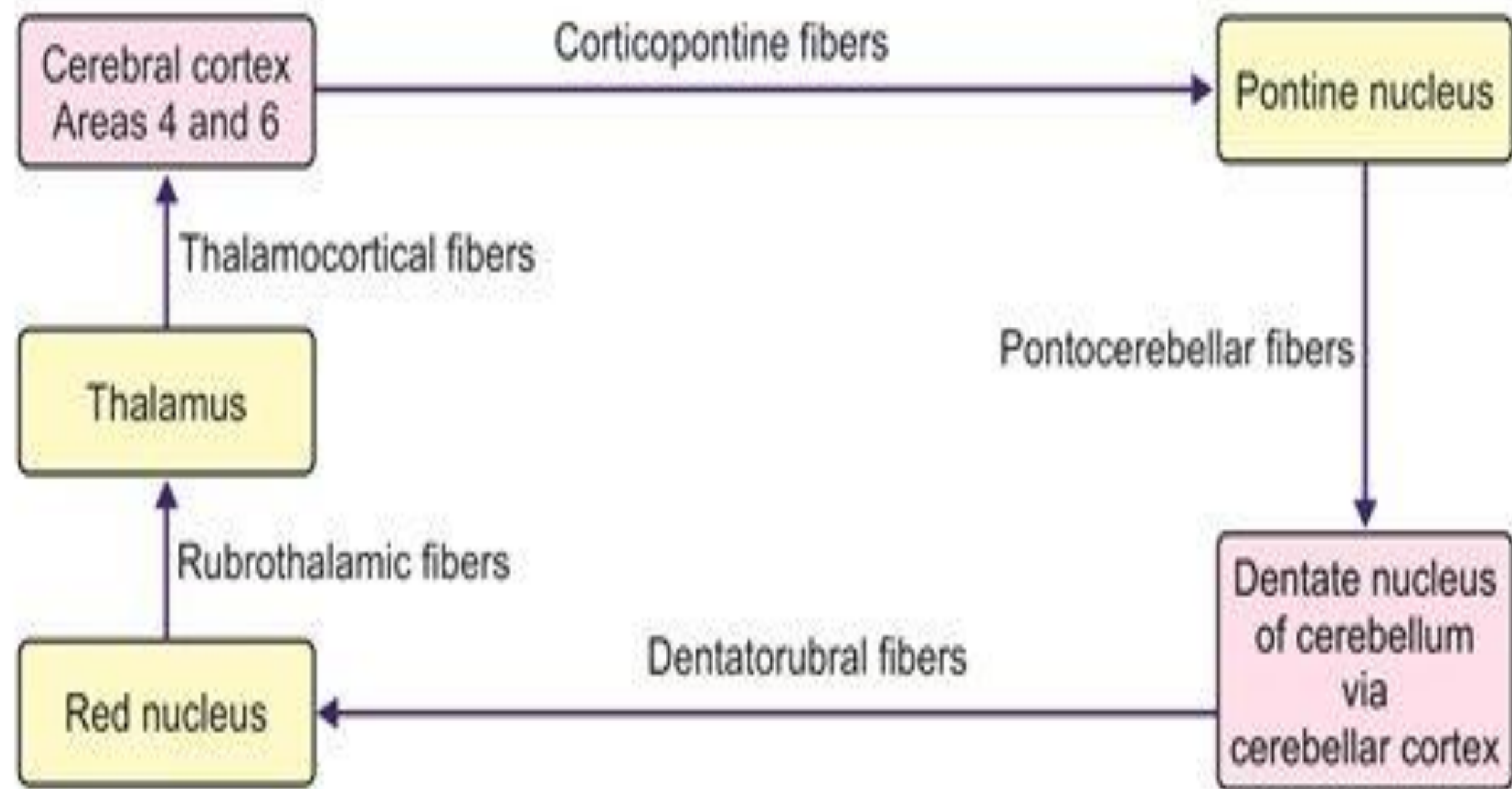
AFFERENT-EFFERENT CIRCUIT

□ Afferent-efferent circuit also called ***cerebro-cerebello-cerebral connection*** is an important neuronal pathway involved in cerebellar control of voluntary movements initiated by motor area of cerebral cortex.

□ This circuit includes two tracts, namely

Cerebropontocerebellar tract: fibers from motor areas 4 & 6 in frontal lobe of cerebral cortex enter the pontine nuclei & pass via middle cerebellar peduncle of opposite side as pontocerebellar fibers to end in the cerebellar cortex.

Dentatorubrothalamocortical tract: fibers from dentate nucleus pass through superior cerebellar peduncle to end in red nucleus of opposite side as dentatorubral fibers & project into thalamus as rubrothalamic fibers which is connected to areas 4 & 6 in motor cortex of cerebrum via thalamocortical fibers.



FUNCTIONS OF CORTICOCEREBELLUM

Damping action

Corticocerebellum receives impulses from motor cortex as well as feedback signals from muscles to prevent exaggerated muscular activity by cutting-off any extra impulses.

Control of ballistic movements

Rapid alternate movements that take place in different parts of body while doing any skilled work like typing, cycling, dancing etc are called ballistic movements.

Timing & programming of movements

While doing any fast-skilled work(e.g. using a typewriter)a chain of movements occur rapidly in a sequential manner.

Corticocerebellum plans these sequential movements by planning schedule of time duration of each movement & time interval between movements.

Cont..

Servomechanism

Servomechanism is the correction of any disturbance or interference while doing skilled work.

Comparator function

- Cerebellum receives information from cerebral cortex regarding cortical impulses sent to muscles as well as receives proprioceptive impulses from muscles regarding their actions.
- By receiving signals from both ends, corticocerebellum compares the cortical commands for muscle activity & the actual movements performed by muscles.
- If any correction is to be done then corticocerebellum send impulses to the motor cortex to correct or modify the commands to muscles.

CEREBELLAR LESIONS

DISTURBANCES IN TONE & POSTURE

Atonia or hypotonia

□ Loss or reduction in muscle tone due to cerebellar lesion causes disturbances in postural reflexes.

Changes in attitude

- Rotation of head towards the unaffected side.
- Lowering of shoulder & abduction(outward movement)of leg on the affected side.
- Weight of body is placed on leg of unaffected side resulting in trunk bent with concavity towards the affected side.

Cont..

Lateral deviation of arms

In unilateral lesion, arm of affected side deviates laterally while in bilateral lesion, both arms deviate laterally when both arms are stretched & held in front of body with closed eyes.

Effect on deep reflexes

Pendular movements occur while eliciting a tendon jerk (especially knee jerk, patellar tendon reflex).

DISTURBANCES IN EQUILIBRIUM

□ Legs are spread to provide a broad base while standing & body sways side to side with oscillations of head.

□ A staggering, reeling (drunken like) walking pattern is observed in cerebellar lesion.

DISTURBANCES IN MOVEMENTS

Ataxia: lack of coordination of movements.

Asynergia: lack of coordination between different muscle groups i.e. protagonists, antagonists, synergists.

Asthenia: weakness, fatigued easily & slowness of muscles.

Hypermetria/hypometria: the arm may overshoot or it may fall short while reaching an object.

Intention tremor: tremor while attempting any voluntary action.

Astasia: unsteady voluntary movements.

Nystagmus: to & fro movement of eyeball.

Rebound phenomenon: when patient attempts to do movement against resistance, the limb moves forcibly in the direction of attempt if resistance is removed suddenly.

Dysarthria: disturbances in speech.

Adiadochokinesis: inability to do rapid alternate successive movements.

POST ASSESSMENT

Q1. The deep cerebellar nucleus associated with vestibulocerebellum is:

- a) Fastigial
- b) Emboliform
- c) Globose
- d) Dentate

Q2. Damage to cerebellum most commonly produce:

- a) Paralysis
- b) Ataxia
- c) Aphasia
- d) hemianopia

Cont..

Q3. The spinocerebellar tract mainly conveys information about:

- a) Vision
- b) Hearing
- c) Taste
- d) Proprioception

Q4. Corticopontocerebellar fibers enter the cerebellum through:

- a) Superior cerebellar peduncle
- b) Middle cerebellar peduncle
- c) Inferior cerebellar peduncle
- d) Medial longitudinal fasciculus